

Speech Signal Processing: BSAI - 7th Sem

Lab 2 Task Report

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Report

1. Unit Impulse Sequence

The unit impulse sequence is a fundamental discrete-time signal, denoted as $\delta[n]$, which has a value of 1 at $n = 0$ and 0 elsewhere. In this task, we define the impulse over the range $n = -10$ to 10. This signal is commonly used in digital signal processing to test or characterize systems, as its convolution with any system yields the system's impulse response. The plotted output shows a single spike at $n = 0$, confirming the correct implementation of the impulse function.

```
black
black1 close all;
black2 clear;
black3 clc
black4 %Unit Impulse
black5 figure(1)
black6 n = -10:10;
black7 impulse = [zeros(1, 10) 1 zeros(1, 10)];
black8 stem(n, impulse);
black9 title('Unit-ImpulseE-Sequence');
black10 xlabel('n');
black11 ylabel('Amplitude');
black12 grid on;
```

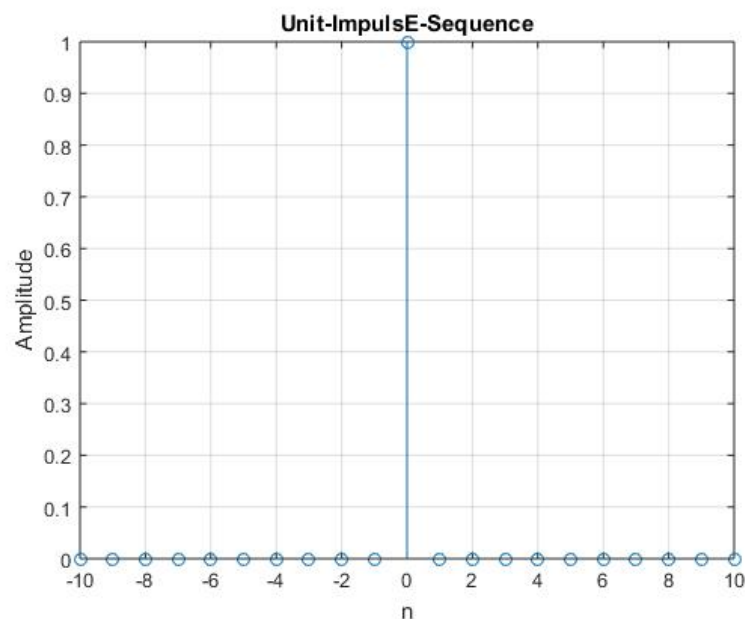


Figure 1: Unit Impulse Sequence

2. Delayed Unit Impulse Sequence

This task demonstrates a delayed version of the unit impulse. We define a delay factor $M = 3$, meaning the non-zero value of the impulse occurs at $n = 3$. The delay is implemented by checking when the index equals M . The resulting plot shows a single spike at $n = 3$ within the range $n = -10$ to 10, assuming M is within this range.

Delayed impulses are important in system analysis and modeling memory behavior in digital systems. If M lies outside the index range, the spike would not be visible.

```

black
black1 close all;
black2 clear;
black3 clc
black4 %Delayed Unit Impulse
black5 figure(2)
black6 n = -10:10;
black7 M = 3;
black8 delayed_impulse = (n == M);
black9 stem(n, delayed_impulse);
black10 title('Delayed-Unit-Impulse-Sequence');
black11 xlabel('n');
black12 ylabel('Amplitude');
black13 grid on;

```

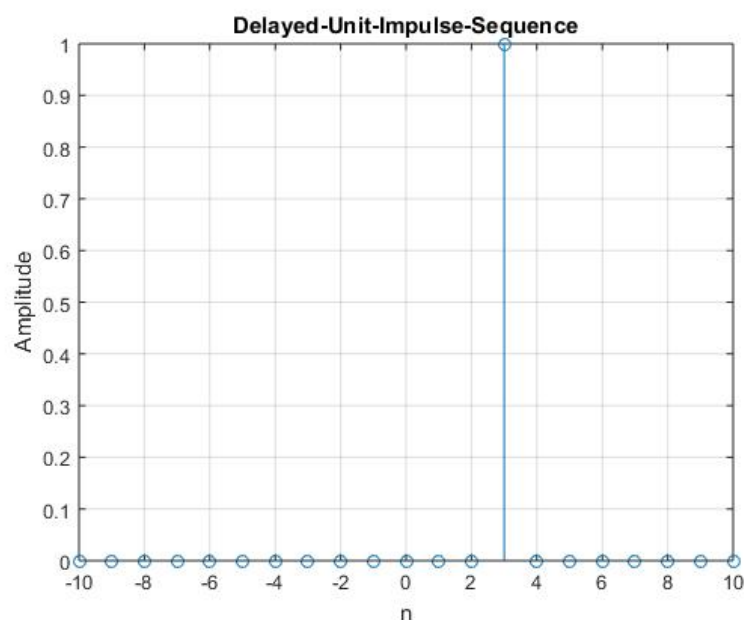


Figure 2: Delayed Unit Impulse Sequence

3. Unit Step Sequence

The unit step function, denoted as $u[n]$, is defined as 0 for $n < 0$ and 1 for $n \geq 0$. In this experiment, the unit step was created by using a threshold condition on the index n . This signal is used in system modeling to represent signals that start at a specific point in time. It helps simulate turning a system "on" at $n = 0$. The plot correctly shows values of zero for negative indices and one for zero and positive indices, demonstrating the step behavior.

```

black
black1 close all;
black2 clear;
black3 clc
black4 %Unit Step

```

```

black5 figure(3)
black6 n = -10:10;
black7 unit_step = (n>=0);
black8 stem(n, unit_step);
black9 title('Unit-Step-Sequence');
black10 xlabel('n');
black11 ylabel('Amplitude');
black12 grid on;

```

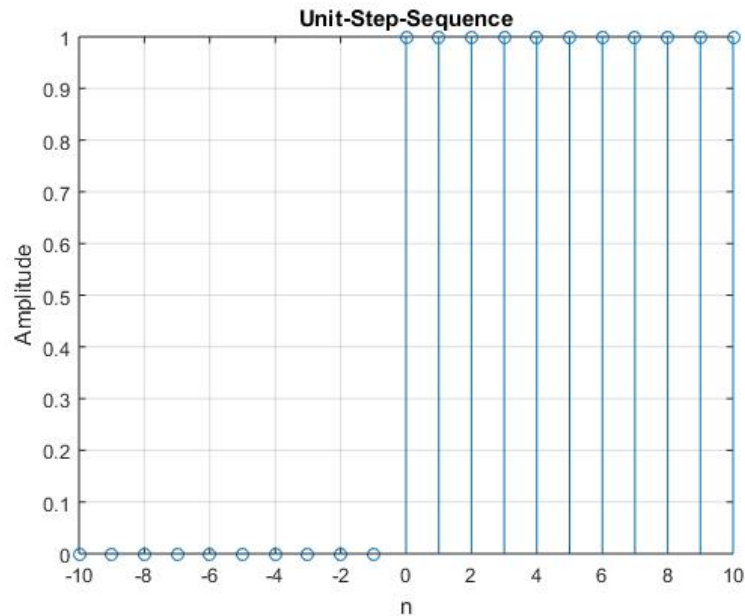


Figure 3: Unit Step Sequence

4. Sinusoidal Sequence

In this task, we generate a discrete-time sinusoidal signal using the equation $x[n] = \sin(2\pi fn)$ with a normalized frequency $f = 0.1$. The sequence is plotted over the range $n = 0$ to 20. The resulting plot shows a smooth periodic waveform, demonstrating the basic characteristics of discrete sinusoidal signals. These waveforms form the building blocks for analyzing signals in the frequency domain using techniques like Fourier Transform. The periodicity and symmetry of the waveform are clearly visible in the plot.

```

black
black1 close all;
black2 clear;
black3 clc
black4 %Sinusoidal Sequences
black5 figure(4)
black6 n = 0:20;
black7 x = sin(2*pi*0.1*n);
black8 stem(n, x);
black9 title('Sinusoidal-Sequences');
black10 xlabel('n');
black11 ylabel('Amplitude');
black12 grid on;

```

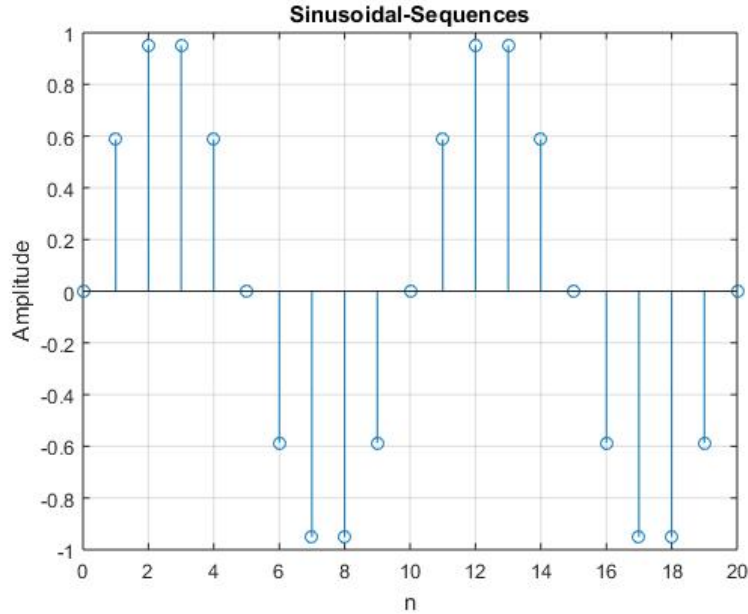


Figure 4: Sinusoidal Sequence

5. Linearity of System

This task investigates the linearity property of a system using the superposition principle. Two input signals $x_1[n]$ and $x_2[n]$ are combined linearly using weights a and b , and passed through a digital filter. If the system is linear, the output due to the weighted input should be equal to the weighted sum of individual outputs. The difference signal $d[n] = y[n] - y_t[n]$ is also plotted to highlight any discrepancy. A negligible difference confirms the system's linear behavior. Linearity is crucial in many DSP applications, as it allows predictable system responses.

```

black
black1 close all;
black2 clear;
black3 clc
black4 %Linear and Nonlinear Systems
black5 figure(5)
black6 n = 0:40;
black7 a = 2;
black8 b = -3;
black9 x1 = cos(2*pi*0.1*n);
black10 x2 = cos(2*pi*0.4*n);
black11 x = a*x1 + b*x2;
black12 num = [2.2403 2.4908 2.2403];
black13 den = [1 -0.4 0.75];
black14 ic = [0 0]; %set zero initial conditions
black15 y1 = filter(num, den, x1, ic); %Compue the output y1[n]
black16 y2 = filter(num, den, x2, ic); %Compue the output y2[n]
black17 y = filter(num, den, x, ic); %Compue the output y[n]
black18 yt = a*y1 + b*y2;
black19 d = y - yt ; %Compue the difference output d[n]
black20 %plot the outputs and difference signal
black21 subplot(3,1,1)
black22 stem(n,y);
black23 ylabel('Amplitude');

```

```

black24 title('Output-Due-to-Weighted-Input');
black25 subplot(3,1,2)
black26 stem(n,yt);
black27 ylabel('Amplitude');
black28 title('Weighted-Output');
black29 subplot(3,1,3)
black30 stem(n,d);
black31 xlabel('Time index n');
black32 ylabel('Amplitude');
black33 title('Difference-Signal');
black34 title('Output-Due-to-Weighted-Input');

```

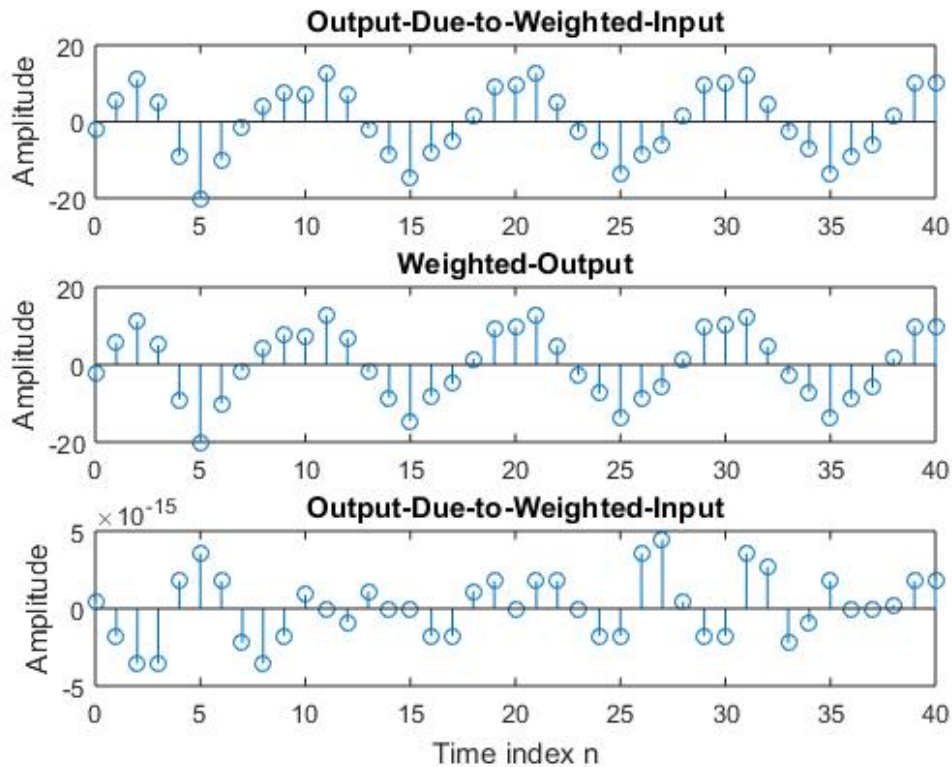


Figure 5: Linearity Test: Output Comparison

6. Time Invariance of System

Time invariance implies that delaying the input results in an identical delay in the output. In this task, the input is delayed by $D = 10$ samples and passed through the same system. The output is compared with a delayed version of the original output. The difference signal highlights whether the system is time-invariant. In this case, the plots show a significant difference between the two, indicating the system is **not** time-invariant. Time invariance is important in system stability and predictability for signal analysis and real-time processing.

```

black
black1 close all;
black2 clear;
black3 clc

```

```

black4 %Time-Invariant and Time-Varying Sytem
black5 figure(6)
black6 n = 0:40;
black7 D = 10;
black8 a = 3.0;
black9 b = -2;
black10 x = a*cos(2*pi*0.1*n) + b*cos(2*pi*0.4*n);
black11 xd = [zeros(1,D) x];
black12 num = [2.2403 2.24908 2.2403];
black13 den = [1 -0.4 0.75];
black14 ic = [0 0]; %Set Initial Condition
black15 %Compute the output y[n]
black16 y = filter(num, den, x, ic);
black17 %Compute the output yd[n]
black18 yd = filter(num, den, xd, ic);
black19 %Compute the difference output d[n]
black20 d = y-yd(1:D:41:D);
black21 %Plot the outputs subplot
black22 subplot(3,1,1)
black23 stem(n,y);
black24 ylabel('Amplitude');
black25 title('Output-y[n]');
black26 grid;
black27 subplot(3,1,2)
black28 stem(n,yd(1:41));
black29 ylabel('Amplitude');
black30 title('Output-Due-to-Delayed-Input');
black31 grid;
black32 subplot(3,1,3)
black33 stem(n,d);
black34 xlabel('Time Index n')
black35 ylabel('Amplitude');
black36 title('Difference Signal');
black37 grid;

```

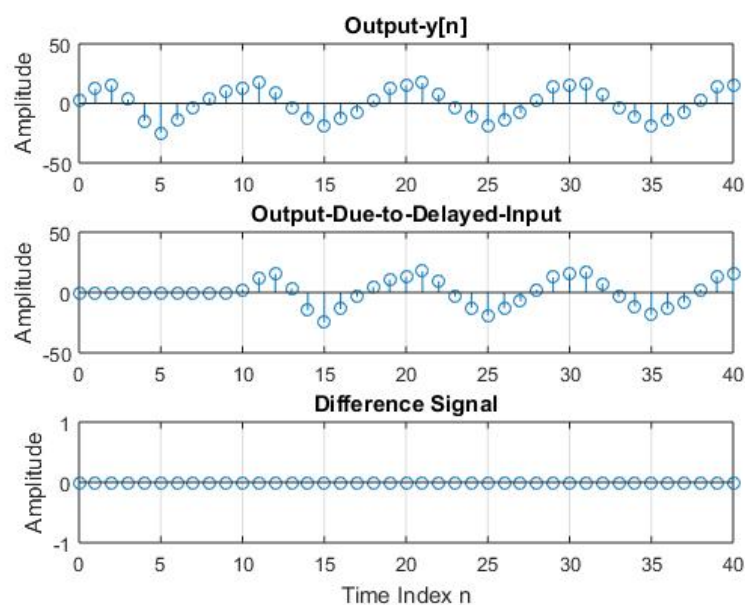


Figure 6: Time Invariance Test

7. Impulse Response of LTI System

The impulse response of a system is its output when an impulse signal is applied. In this task, MATLAB's `impz` function is used to compute the impulse response of the system described by its numerator and denominator coefficients. The response is computed over 400 samples. The resulting plot shows how the system reacts to a unit impulse, which is critical in characterizing Linear Time-Invariant (LTI) systems. The impulse response provides complete information about the system's behavior and is used extensively in convolution and frequency analysis.

```
black
black1 close all;
black2 clear;
black3 clc
black4 %Linear Time-Invariant Discrete-Time Systems
black5 figure(7)
black6 %Compute the impulse response
black7 N = 10;
black8 num = [2.2403 2.4908 2.2403];
black9 den = [1 -0.4 0.75];
black10 y = impz(num,den,N);
black11 %Plot the impulse response
black12 stem(y);
black13 xlabel('Time index n');
black14 ylabel('Amplitude');
black15 title('Impulse Response');
black16 grid
```

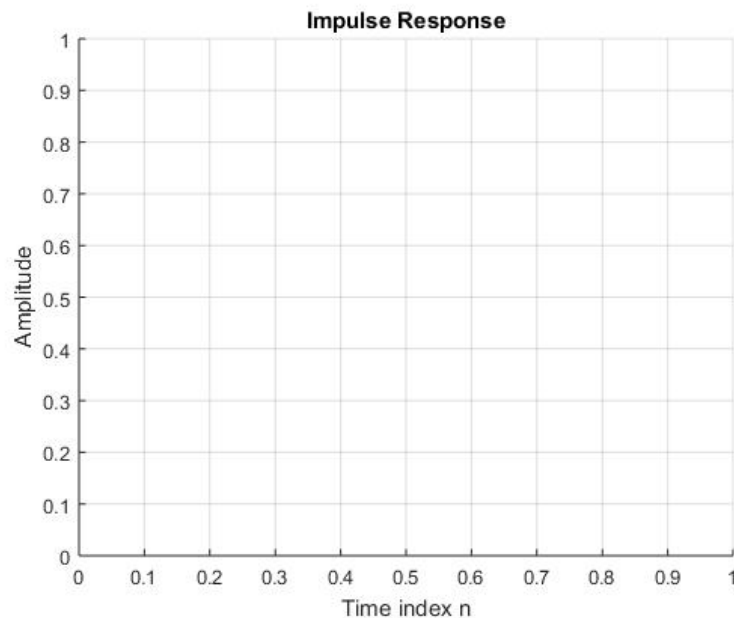


Figure 7: Impulse Response of LTI System